MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, VHSIC, CMOS, 65,536-BIT SELECTABLE MODE, STATIC RANDOM ACCESS MEMORY (SRAM), MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS, 65,536-bit selectable operating mode, static random access memory microcircuits. These microcircuits conform to the functional throughput rate as defined in the phase I very high speed integrated circuit (VHSIC) program. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit organization	Access time
01	16,384 words x 4 bits	35 ns
02	16,384 words x 4 bits	45 ns
03	16,384 words x 4 bits	55 ns

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-STD-883.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
X	Figure 1 (22-lead, 1.115" x .320" x .170"), ceramic side brazed dual-in-line package

1.3 Absolute maximum ratings.

Supply voltage (V_{CC})	-0.5 V dc to +7.0 V dc
All input or output voltages	-0.5 V dc to +7.0 V dc
DC output current	20 mA
Storage temperature range	-65°C to +150°C
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC})	15°C/W
Maximum power dissipation (P_D)	1.0 W
Maximum junction temperature (T_J) 1/	175°C

1/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc to 5.5 V dc
Ground (V_{SS})	0.0 V dc
High level input voltage (V_{IH})	2.2 V dc to V_{CC}
Low level input voltage (V_{IL})	-0.5 V dc to 0.8 V dc
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1A.

3.2.2 Functional block diagram. The functional block diagram shall be as specified on figure 2. Upon implementing design changes, the manufacturer must submit a new block diagram to the qualifying activity for inclusion in this specification. The block diagram shall clearly define the row address inputs and the column address inputs.

3.2.3 Truth table. The truth table shall be as specified on figure 3.

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3.2.4 Functional tests. The functional tests used to test this device are contained in the appendix. If the test patterns cannot be implemented due to test equipment limitations, then alternate test patterns to accomplish the same results shall be submitted to the qualifying activity for approval.

3.2.4.1 Functional test devices. Devices which fail a maximum of five bits during functional testing may be used for destructive quality conformance inspection testing (QCI) provided subject parts are identified as NONSHIPPABLE parts and the failed bits are identified through the remainder of testing. (The devices defined herein shall be for group C and D end-points only.)

3.2.5 Case outline. The case outline shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II herein. The electrical tests for each subgroup are described in table III herein and in table V test vectors that forms a part of the manufacturing test tape.

3.5.1 Behavorial (VHDL) model. The electrical behavior of this device shall be documented in VHSIC Hardware Description Language (VHDL), IEEE 1076-1987. This documentation shall be submitted for DESC approval.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Correctness of indexing and marking. All devices shall be subject to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally an approved electrical test may be devised especially for this requirement.

3.6.2 Total dose radiation hardness identifier. The total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and herein.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 41 (see MIL-M-38510, appendix E).

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TABLE I. Electrical performance characteristics. 1/

Parameter	Symbol	Conditions V_{SS} = 0.0 V; V_{CC} = 5.0 V ±10% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Device type	Limits		Unit
				Min	Max	
High level input voltage	V_{IH}		All	2.2		V
Low level input voltage	V_{IL}		All		0.8	V
High level output voltage	V_{OH}	V_{CC} = 4.5 V I_{OH} = -4 mA	All	2.4		V
Low level output voltage	V_{OL}	V_{CC} = 5.5 V I_{OL} = 8 mA	All		0.4	V
Output leakage current	I_{OZH}	V_O = 5.5 V; $\overline{CS}$, $\overline{WE}$ = 2.2 V; V_{CC} = 5.5 V	All		+2	µA
	I_{OZL}	V_O = 0.0 V; $\overline{CS}$, $\overline{WE}$ = 2.2 V; V_{CC} = 5.5 V			-2	
High level input current	I_{IH}	Input under test = 5.5 V All other inputs = 0.0 V	All		+2	µA
Low level input current	I_{IL}	Input under test = 0.0 V All other inputs = 5.5 V			-2	µA
Supply current (standby)	I_{CCSB}	$\overline{CS}$ = 4.5 V, $\overline{WE}$ = 0.0 V A0-A13 = 5.5 V	All		15	mA
Supply current (operating)	I_{CC}	$\overline{CS}$, $\overline{WE}$ = 0.0 V A0-A13 = 5.5 V I/O1-I/O4 = open	All		100	mA
Input capacitance	C_I	V_I = 0.0 V, f = 1 MHz V_{CC} = 5.5 V	All		5	pF
Output capacitance	C_O	V_O = 0.0 V, f = 1 MHz V_{CC} = 5.5 V	All		7	pF

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued. 1/

Parameter	Symbol	Conditions V_{SS} = 0.0 V; V_{CC} = 5.0 V ±10% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Address access time	t_{AVQV}	See figure 5	01 02 03		35 45 55	ns
Chip select access time	t_{ELQV}		01 02 03		35 45 55	ns
Chip select to output active	t_{ELQX}		All	0		ns
Chip select to output disable (High-Z)	t_{EHQZ}		01 02 03		15 20 25	ns
Output hold from address change	t_{AVQX}		All	5		ns
Chip disable to power down delay	t_{PD}		01 02 03		30 40 50	ns
Chip enable to power up delay	t_{PU}		All	0		ns
Chip select to end of write	t_{ELWH}		01 02 03	30 35 45		ns
Address setup time	t_{AVWL}		All	0		ns
Address valid to end of write	t_{AVWH}		01 02 03	30 35 45		ns
Write pulse width	t_{WLWH}		01 02 03	30 35 45		ns
Write recovery time	t_{WHAX}		All	0		ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued. 1/

Parameter	Symbol	Conditions V_{SS} = 0.0 V; V_{CC} = 5.0 V ±10% -55°C ≤ T_C ≤ +125°C unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Write enable low to output in High-Z	t_{WLQZ}		01,02		15	ns
			03		20	
Data setup to end of write	t_{DVWH}		01 02 03	15 20 25		ns
Data hold after end of write	t_{WHDX}		All	0		ns
Write enable high to data out active time	t_{WHQX}		All	0		ns

1/ Complete terminal conditions shall be specified in table III.

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TABLE II. Burn-in and electrical test requirements. 1/ 2/ 3/ 4/

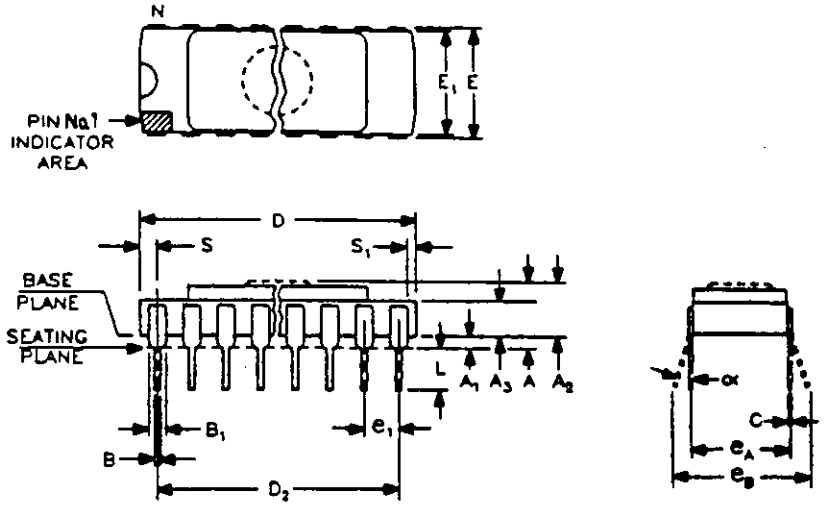
Line no.	Applicable tests and MIL-STD-883 test method	Class S device		Class B devices	
		Table III subgroups 2/	Table IV delta 3/ limits	Table III subgroups 2/	Table IV delta 3/ limits
1	Interim electrical parameters (method 5004)	2, 10		2, 10	
2	Static burn-in I (method 1015)	Required		Not required	
3	Same as line 1	*2, 10	Δ		
4	Static burn-in II (method 1015)	Required		Not required	
5	Same as line 1	*2, 10	Δ		
6	Dynamic burn-in (method 1015)	Required		Required 5/	
7	Same as line 1	*2, 10	Δ		Δ
8	Final electrical parameters (method 5004)	1,*2,3,7, 8,9,10,11		1,*2,3,7, 8,9,10,11 6/	
9	Group A test requirements (method 5005)	1,2,3,4,7 7/, 8,9,10,11		1,2,3,4,7 7/, 8,9,10,11	
10	Group B end-point electrical parameters (method 5005)	1† , 9, 10,11 4/	Δ	1 † 4/	
11	Group C end-point electrical parameters (method 5005)			2, 10	Δ
12	Group D end-point electrical parameters (method 5005)	2,8(125°C), 10		2, 10	

1/ Blank spaces indicate tests are not applicable.
2/ * indicates PDA applies to subgroup 2 (see 4.2.1).
3/ Δ indicates delta limit shall be required only on table III, subgroup 2 where specified, and the delta limits shall be computed with reference to the previous interim electrical parameters (line 1).
4/ † indicates applies to electrostatic discharge sensitivity tests.
5/ The device manufacturer may at his option, either complete subgroup 2 electrical parameter measurements, including delta measurements within 96 hours after burn-in completion (removal of bias); or may complete subgroups 1 and 7 electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias).
6/ Where the 24 hour option in 5/ above is exercised, subgroups 1 and 7 shall be used for PDA calculations.
7/ For subgroup 4 (see 4.4.1c).

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Case X

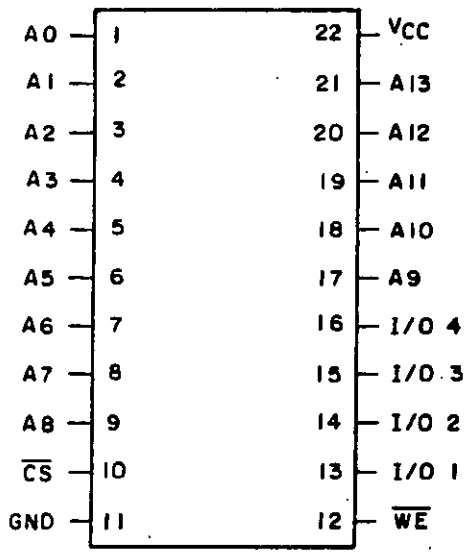
22 lead ceramic in line package intel type C



Family: Ceramic Side Braze Dual In-Line						
Symbol	Millimeters			Inches		
	Min	Max	Notes	Min	Max	Notes
α	0°	10°		0°	10°	
A	2.79	4.32		0.110	0.170	
A_1	0.64	1.27		0.025	0.050	
A_2	2.16	3.40		0.085	0.134	
A_3	2.03	3.05		0.080	0.120	
B	0.38	0.56		0.015	0.022	
B_1	1.27		Typical	0.050		Typical
C	0.23	0.30	Typical	0.009	0.012	Typical
D	27.56	28.32		1.085	1.115	
D_2	25.40		Reference	1.000		Reference
E	7.62	8.13		0.300	0.320	
E_1	7.37	7.62		0.290	0.300	
e_1	2.29	2.79		0.090	0.110	
e_A	7.37		Reference	0.290		Reference
e_B	7.62	9.14		0.300	0.360	
L	3.18	4.06		0.125	0.160	
N	22			22		
S	0.76	1.78		0.030	0.070	
S_1	0.13			0.005		
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FIGURE 1. Case outline.

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Case X

FIGURE 1A. Terminal connections (all device types).

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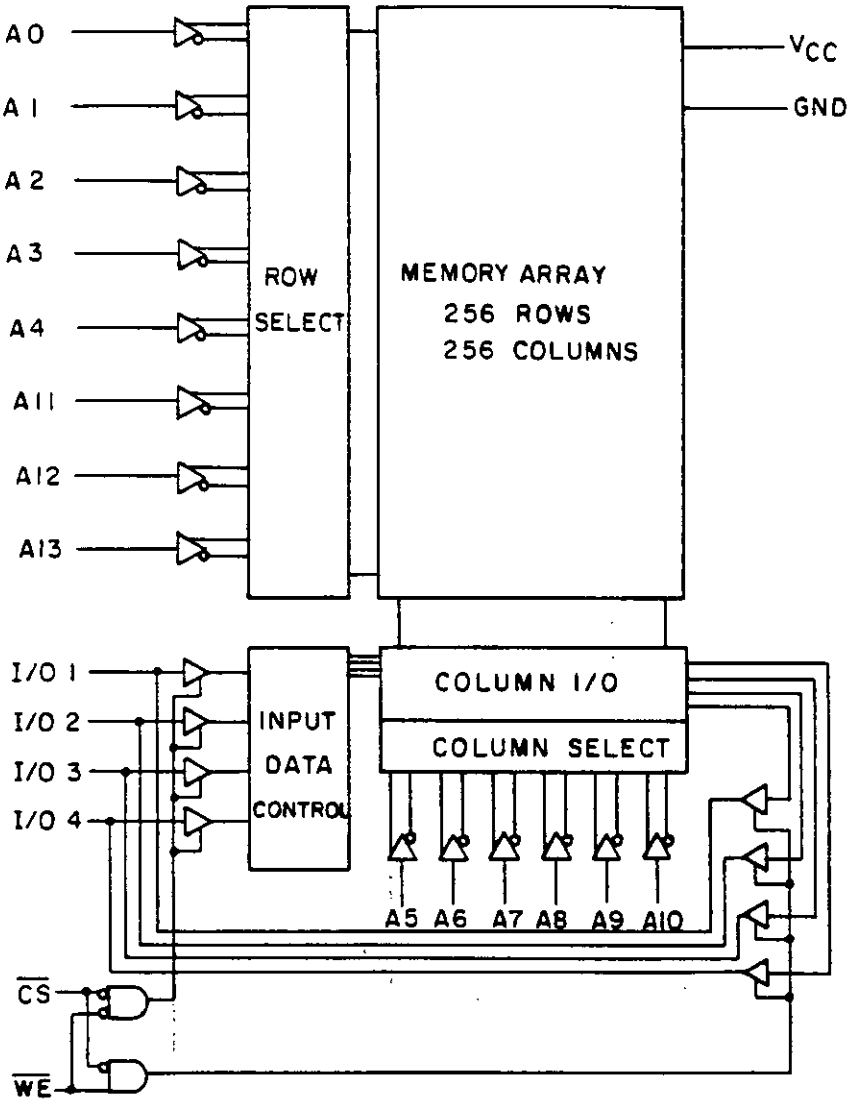


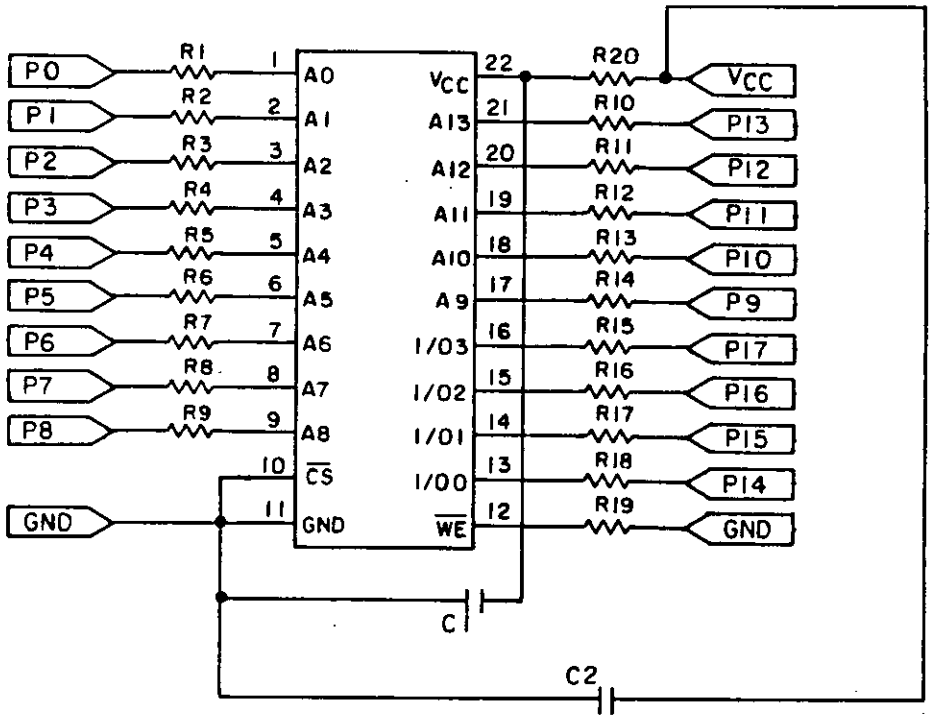
FIGURE 2. Functional block diagram (all device types).

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$\overline{CS}$	$\overline{WE}$	Mode	I/O	Power
H	X	not selected	high-Z	standby
L	L	write	D_{IN}	active
L	H	read	D_{OUT}	active

FIGURE 3. Truth table (all device types).

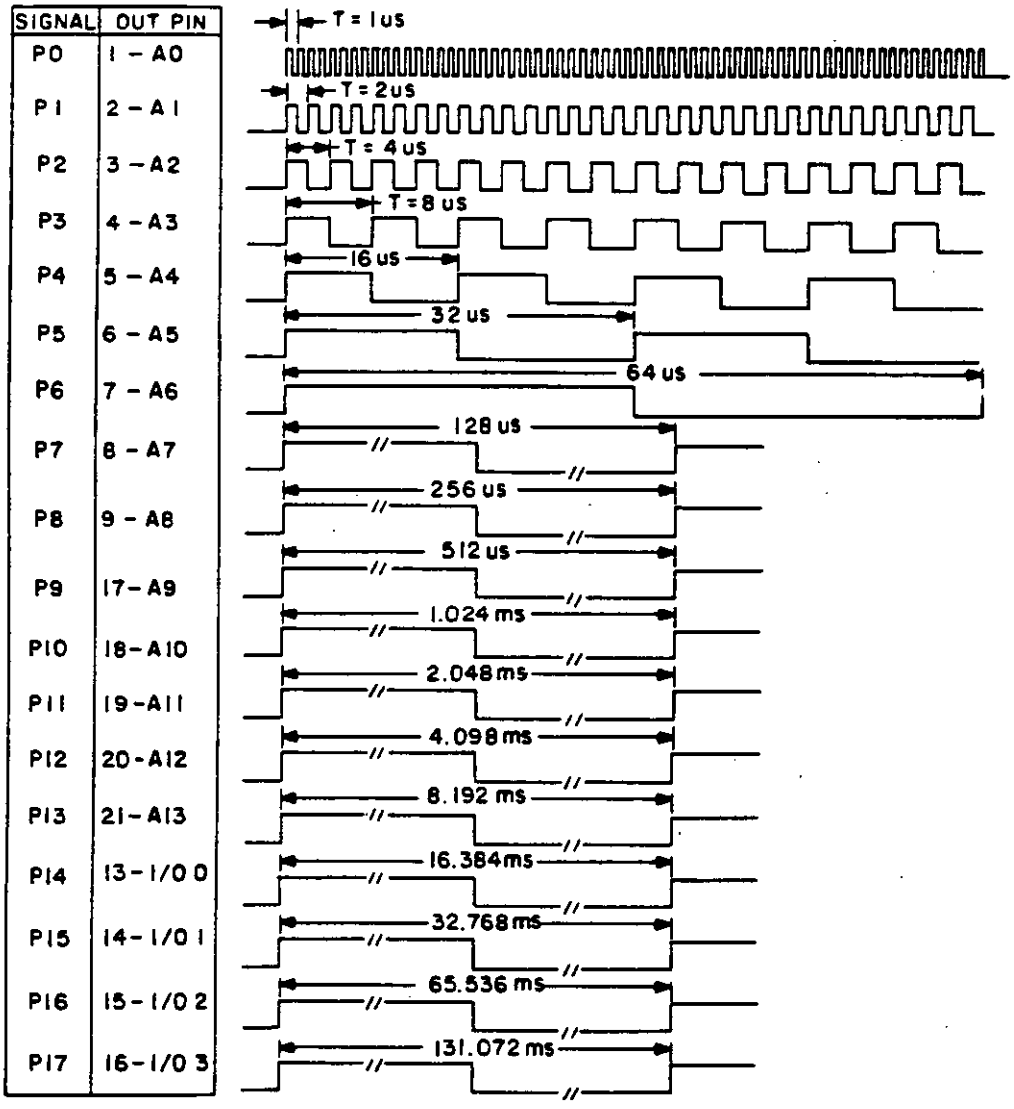
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NOTES:
1. C1 = .1 µF glass encapsulated capacitor (per device).
2. C2 = 47 µF capacitor (one per row).
3. R20 = 10Ω 1% (per device).
4. R1 thru R19 are 1 k 1% metalfilm (per device).
5. ⬠ = input signal.

FIGURE 4. Dynamic burn-in (all device types).

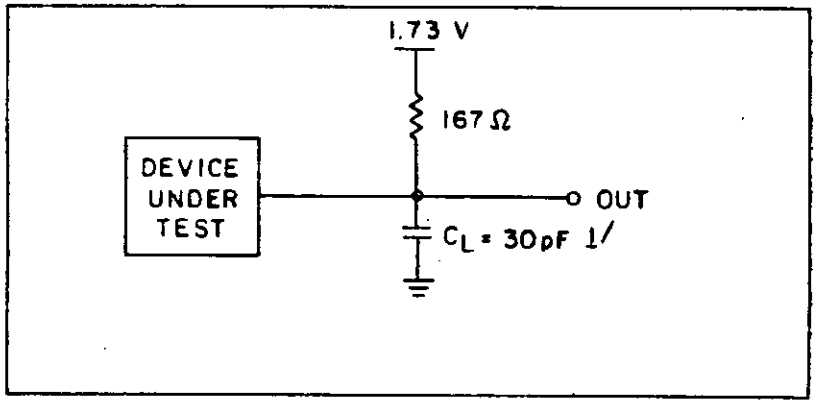
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Signal	VOH	Tolerance	VOL	Tolerance	Timing TOL
P0 - P17	+5.0 Vdc	+/- .25 V	0 V	+/- .25 V	+/- 10%

FIGURE 4. Dynamic burn-in (all device types) - Continued.

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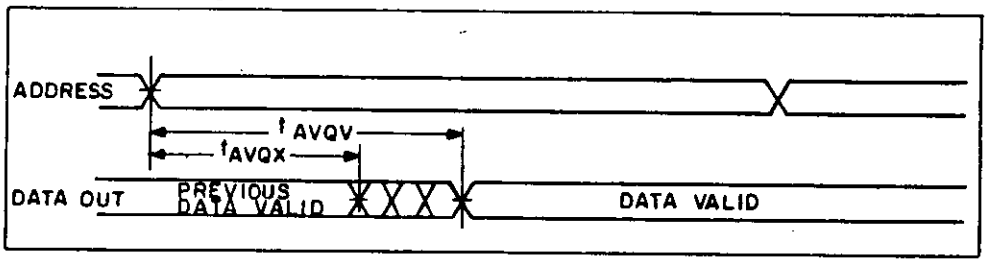


1/ When testing parameters t_{HZ}, t_{LZ}, t_{WZ} and t_{OW} a 5 pf capacitor must be used for C_L in the output load circuit.

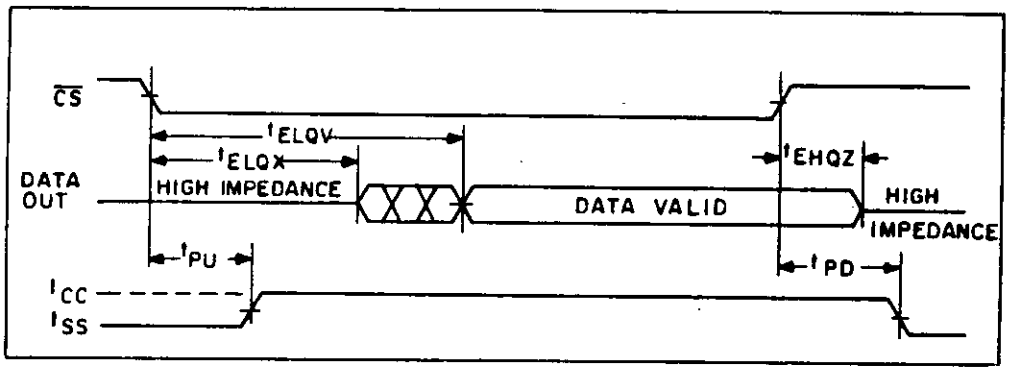
FIGURE 5. Switching time load circuits, test conditions, and waveforms (all device types).

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READ CYCLE NO. 1



READ CYCLE NO. 2

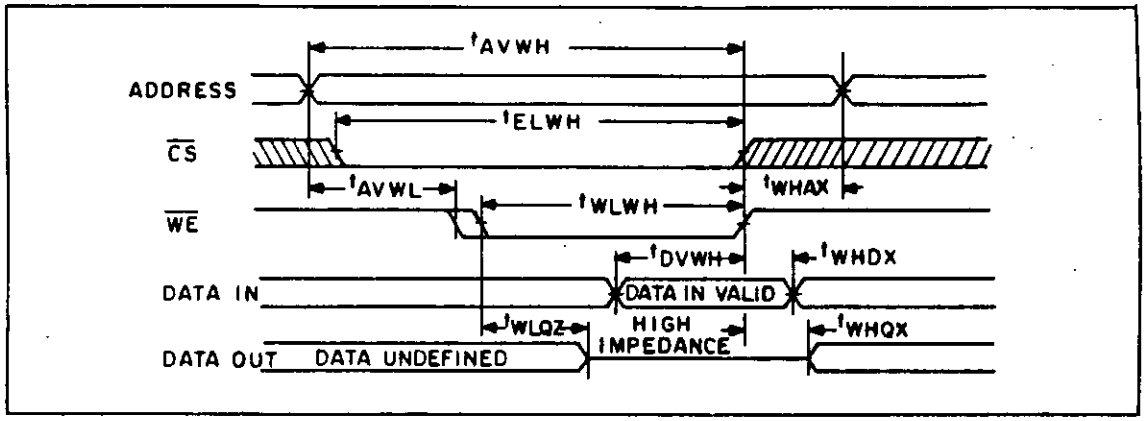


NOTES ON READ OPERATION:
1. All read cycle timings are referenced from the last valid address to the first transitioning address.
2. At any given temperature and voltage condition, t_{EHQZ} maximum is less than t_{ELQX} minimum both for a given device and from device to device.
3. Transition is measured ±500 mV from steady-state voltage t_{ELQX}, t_{EHQZ} only).
4. $\overline{WE}$ is high for read cycles.
5. Device is continously selected, $\overline{CS}$ = V_{IL}.
6. Addresses valid prior to or coincident with $\overline{CS}$ transition low.
7. Case temperatures are "instant on".

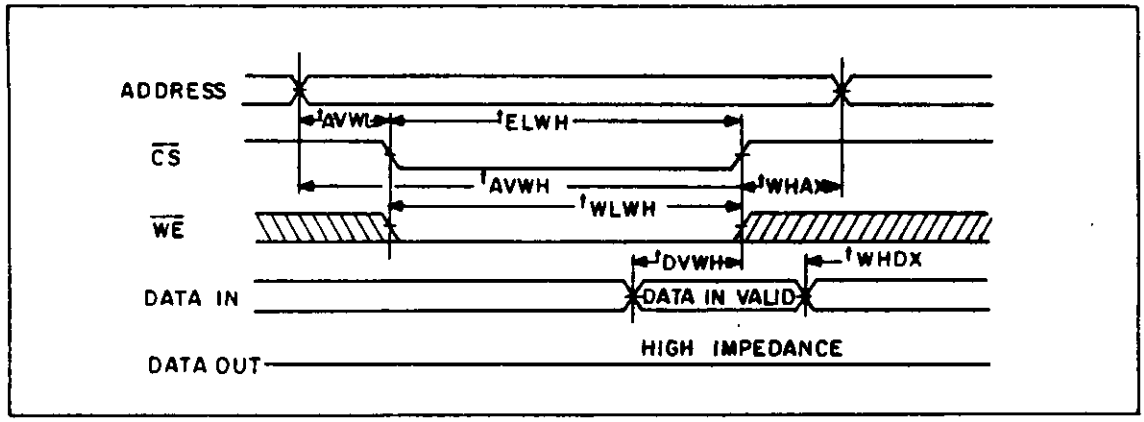
FIGURE 5. Switching time load circuits, test conditions, and waveforms (all device types) - Continued.

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WRITE CYCLE NO. 1 ($\overline{WE}$ CONTROLLED)



WRITE CYCLE NO. 2 ($\overline{CS}$ CONTROLLED)



NOTES ON WRITE OPERATION:
1. All write cycle timings are referenced from the last valid address to the first transitioning address.
2. Transition is measured ±500 mV from steady-state voltage t_{WHQX}, t_{WLQZ} only).
3. $\overline{CS}$ or $\overline{WE}$ must be high during address transitions.
4. If $\overline{CS}$ switches low coincident with or after $\overline{WE}$ switches low, the outputs will stay in a high impedance state.
5. If $\overline{CS}$ switches high coincident with or after $\overline{WE}$ switches high, the outputs will stay in a high impedance state.
6. Case temperatures are "instant on".
7. In a $\overline{WE}$-controlled write cycle, voltage on the I/O pins should not fall between V_{IH} minimum and V_{IL} maximum during t_{WLQZ} time. It is recommended that pullup resistors be utilized on the I/O lines.

FIGURE 5. Switching time load circuits, test conditions, and waveforms (all device types) - Continued.

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TABLE III. Group A inspection (all device types).

Subgroup	Symbol	MIL-STD-883-method	Case M	Adapter pin numbers												
				1	2	3	4	5	6	7	8	9	10	11	12	13
			Test no.	A0	A1	A2	A3	A4	A5	A6	A7	A8	CS	GND	ME	I/O1
1 T_C = +125°C	I_{IH}	3010	1	5.5 V	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	
			2	GND	5.5 V	GND	"	"	"	"	"	"	"	"	"	
			3	"	GND	5.5 V	"	"	"	"	"	"	"	"	"	
			4	"	"	GND	5.5 V	"	"	"	"	"	"	"	"	
			5	"	"	"	GND	5.5 V	"	"	"	"	"	"	"	
			6	"	"	"	"	GND	5.5 V	"	"	"	"	"	"	
			7	"	"	"	"	"	GND	5.5 V	"	"	"	"	"	
			8	"	"	"	"	"	"	GND	5.5 V	"	"	"	"	
			9	"	"	"	"	"	"	"	GND	5.5 V	"	"	"	
			10	"	"	"	"	"	"	"	"	GND	5.5 V	"	"	
			11	"	"	"	"	"	"	"	"	"	GND	"	5.5 V	
			12	"	"	"	"	"	"	"	"	"	"	"	GND	
			13	"	"	"	"	"	"	"	"	"	"	"	"	
			14	"	"	"	"	"	"	"	"	"	"	"	"	
			15	"	"	"	"	"	"	"	"	"	"	"	"	
			16	"	"	"	"	"	"	"	"	"	"	"	"	
	I_{IL}	3010	17	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	
			18	5.5 V	GND	5.5 V	"	"	"	"	"	"	"	"	"	
			19	"	5.5 V	GND	"	"	"	"	"	"	"	"	"	
			20	"	"	5.5 V	GND	"	"	"	"	"	"	"	"	
			21	"	"	"	5.5 V	GND	"	"	"	"	"	"	"	
			22	"	"	"	"	5.5 V	GND	"	"	"	"	"	"	
			23	"	"	"	"	"	5.5 V	GND	"	"	"	"	"	
			24	"	"	"	"	"	"	5.5 V	GND	"	"	"	"	
			25	"	"	"	"	"	"	"	5.5 V	GND	"	"	"	
			26	"	"	"	"	"	"	"	"	5.5 V	GND	"	"	
			27	"	"	"	"	"	"	"	"	"	5.5 V	"	GND	
			28	"	"	"	"	"	"	"	"	"	"	"	5.5 V	
			29	"	"	"	"	"	"	"	"	"	"	"	"	
			30	"	"	"	"	"	"	"	"	"	"	"	"	
			31	"	"	"	"	"	"	"	"	"	"	"	"	
			32	"	"	"	"	"	"	"	"	"	"	"	"	
	I_{OZH}		33										1/		1/	5.5 V
			34										"		"	
			35										"		"	
			36										"		"	
	I_{OZL}		37										"		"	GND
			38										"		"	
			39										"		"	
			40										"		"	
	I_{CC}	3005	41	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.0 V	"	0.0 V	
	I_{SB}	3005	42	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	"	
	I_{SB}	3005	43	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	
	V_{OH}	3006	44	2/	2/	2/	2/	2/	2/	2/	2/	2/	2/	"	2/	-4 mA
			45	"	"	"	"	"	"	"	"	"	"	"	"	
			46	"	"	"	"	"	"	"	"	"	"	"	"	
			47	"	"	"	"	"	"	"	"	"	"	"	"	
	V_{OL}	3007	48	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	"	3/	8 mA
			49	"	"	"	"	"	"	"	"	"	"	"	"	
			50	"	"	"	"	"	"	"	"	"	"	"	"	
			51	"	"	"	"	"	"	"	"	"	"	"	"	
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C															
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C															
4 T_C = +25°C	C_I	3012	52	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/
	C_O	3012	53	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/

See footnotes at end of table III.

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TABLE III. Group A Inspection (all device types) - Continued.

Subgroup	Symbol	MIL-STD-883-method	Case W	Adapter pin numbers									Measured terminal	Test limits		Unit
				14	15	16	17	18	19	20	21	22				
			Test no.	I/02	I/03	I/04	A9	A10	A11	A12	A13	V_{CC}		Min	Max	
1 T_C = +125°C	I_{IH}	3010	1				GND	GND	GND	GND	GND	5.5 V	A0		+2	µA
			2				"	"	"	"	"	"	A1		"	"
			3				"	"	"	"	"	"	A2		"	"
			4				"	"	"	"	"	"	A3		"	"
			5				"	"	"	"	"	"	A4		"	"
			6				"	"	"	"	"	"	A5		"	"
			7				"	"	"	"	"	"	A6		"	"
			8				"	"	"	"	"	"	A7		"	"
			9				"	"	"	"	"	"	A8		"	"
			10				"	"	"	"	"	"	CS		"	"
			11				"	"	"	"	"	"	WE		"	"
			12				5.5 V	"	"	"	"	"	A9		"	"
			13				GND	5.5 V	"	"	"	"	A10		"	"
			14				"	GND	5.5 V	"	"	"	A11		"	"
			15				"	"	GND	5.5 V	"	"	A12		"	"
			16				"	"	GND	GND	5.5 V	"	A13		"	"
	I_{IL}	3010	17				5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A0		-2	"
			18				"	"	"	"	"	"	A1		"	"
			19				"	"	"	"	"	"	A2		"	"
			20				"	"	"	"	"	"	A3		"	"
			21				"	"	"	"	"	"	A4		"	"
			22				"	"	"	"	"	"	A5		"	"
			23				"	"	"	"	"	"	A6		"	"
			24				"	"	"	"	"	"	A7		"	"
			25				"	"	"	"	"	"	A8		"	"
			26				"	"	"	"	"	"	CS		"	"
			27				"	"	"	"	"	"	WE		"	"
			28				GND	"	"	"	"	"	A9		"	"
			29				5.5 V	GND	"	"	"	"	A10		"	"
			30				"	5.5 V	GND	"	"	"	A11		"	"
			31				"	"	5.5 V	GND	"	"	A12		"	"
			32				"	"	5.5 V	5.5 V	GND	"	A13		"	"
	I_{OZH}		33	5.5 V								"	I/01		+2	"
			34									"	I/02		"	"
			35		5.5 V							"	I/03		"	"
			36			5.5 V						"	I/04		"	"
	I_{OZL}		37	GND								"	I/01		-2	"
			38									"	I/02		"	"
			39		GND							"	I/03		"	"
			40			GND						"	I/04		"	"
	I_{CC}	3005	41				5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	V_{CC}		100	mA
	I_{SB}	3005	42				4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	V_{CC}		15	"
	I_{SB}	3005	43				5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	V_{CC}		15	"
	V_{OH}	3006	44	-4 mA			2/	2/	2/	2/	2/	4.5 V	I/01	2.4		V
			45				"	"	"	"	"	"	I/02	"		"
			46		-4 mA		"	"	"	"	"	"	I/03	"		"
			47			-4 mA	"	"	"	"	"	"	I/04	"		"
	V_{OL}	3007	48	8 mA			3/	3/	3/	3/	3/	5.5 V	I/01		0.4	"
			49				"	"	"	"	"	"	I/02		"	"
			50		8 mA		"	"	"	"	"	"	I/03		"	"
			51			8 mA	"	"	"	"	"	"	I/04		"	"
2																
3																
4 T_C = +25°C	C_I	3012	52	4/	4/	4/	4/	4/	4/	4/	4/	4/	Inputs	5		pF
	C_O	3012	53	4/	4/	4/	4/	4/	4/	4/	4/	4/	Outputs	7		pF

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TABLE III. Group A Inspection (all device types) - Continued.

Subgroup	Symbol	MIL-STD-883-method	Case W	Adapter pin numbers												
				1	2	3	4	5	6	7	8	9	10	11	12	13
			Test no.	A0	A1	A2	A3	A4	A5	A6	A7	A8	$\overline{CS}$	GND	$\overline{WE}$	I/O1
T_C = 7 +25°C	Functional test	3014	54	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/	5/
8A	Same tests, terminal conditions, and limits as subgroup 7, except T_C = +125°C															
8B	Same tests, terminal conditions, and limits as subgroup 7, except T_C = -55°C															
T_C = 9 +25°C	t_{AVQX}		55	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/	6/ 7/
	t_{ELQX}		56	8/	8/	8/	8/	8/	8/	8/	8/	8/	8/	8/	8/	7/	8/
	t_{AVWL}		57	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	9/	"	9/
	t_{WHAX}		58	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	"	11/
	t_{WHDX}		59	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	"	11/
	t_{WHQX}		60	10/	10/	10/	10/	10/	10/	10/	10/	10/	10/	10/	10/	"	10/
	t_{AVQV}		61	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	"	11/
	t_{ELQV}		62	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	"	11/
	t_{EHQZ}		63	8/	8/	8/	8/	8/	8/	8/	8/	8/	8/	8/	8/	"	8/
	t_{ELWH}		64	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	11/	"	11/
	t_{AVWH}		65	"	"	"	"	"	"	"	"	"	"	"	"	"	"
	t_{WLWH}		66	"	"	"	"	"	"	"	"	"	"	"	"	"	"
	t_{DVWH}		67	"	"	"	"	"	"	"	"	"	"	"	"	"	"
	t_{WLQZ}		68	10/	10/	10/	10/	10/	10/	10/	10/	10/	10/	10/	10/	"	10/
10	Same tests, terminal conditions, and limits as subgroup 9, except T_C = +125°C															
11	Same tests, terminal conditions, and limits as subgroup 9, except T_C = -55°C															

1/ Terminal conditions for the output leakage current tests shall be as follows:

a. V_{IH} = 2.2 V, V_{IL} = 0.8 V.

b. For I_{OZH}: Apply V_{IH} to $\overline{CS}$ and $\overline{WE}$. Measure the leakage current on the output pins while applying the specified voltage.

c. For I_{OZL}: Apply V_{IH} to $\overline{CS}$ and $\overline{WE}$. Measure the leakage current on the output pins while applying the specified voltage.

2/ An input preconditioning sequence shall be applied that results in a logic "1" at the output to be measured. Logic input levels for address pins are V_{IL} = 0.8 V and V_{IH} = 2.2 V. CS is at 0.8 V; WE is at 2.2 V.

3/ An input preconditioning sequence shall be applied that results in a logic "0" at the output to be measured. Logic input levels for address pins are V_{IL} = 0.8 V and V_{IH} = 2.2 V. CS is at 0.8 V; WE is at 2.2 V.

4/ Input/output capacitance shall be measured between the designated terminal and the GND pin under the following conditions: V_I = 0.0 V, f = 1 MHz, oscillator voltage = 50 mV rms maximum. Unused pins are open. One pin of each input/output driver type shall be tested on each sample device. Testing is done for initial qualification and after process or design changes which may affect input and/or output capacitance.

5/ The functional tests in subgroups 7 and 8 shall verify the applicable test patterns, see table VI herein. All bits shall be tested. Terminal conditions are as follows:

a. Inputs: H = 3.0 V, L = 0.0 V.

b. Outputs: H > 1.5 V, L < 1.5 V.

c. The functional tests shall be performed with V_{CC} = 4.5 V and V_{CC} = 5.5 V.

6/ For this test in subgroups 9, 10, and 11, the LDRCOL pattern, see table VI shall be used. Input/output conditions appear in table IIIA. Timing diagrams appear in the write and read cycles of figure 5. Subgroups 9, 10, and 11 shall be performed with V_{CC} = 4.5 V and 5.5 V.

7/ The outputs are loaded in accordance with figure 5.

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TABLE III. Group A Inspection (all device types) - Continued.

Subgroup	Symbol	MIL-STD-883-method	Case W	Adapter pin numbers									Measured terminal	Test limits		Unit
				14	15	16	17	18	19	20	21	22				
			Test no.	1/02	1/03	1/04	A9	A10	A11	A12	A13	V_{CC}		Min	Max	
7 T_C = +25°C	Functional test	3014	54	5/	5/	5/	5/	5/	5/	5/	5/	5/	1/01 to 1/04	5/		
8A																
8B																
9 T_C = +25°C	t_{AVQX}		55	6/ 7/	6/ 7/	6/ 7/	6/	6/	6/	6/	6/	6/	1/01 through 1/04	5		ns
	t_{ELQX}		56	7/ 8/	7/ 8/	7/ 8/	8/	8/	8/	8/	8/	8/		0		"
	t_{AVWL}		57	" 9/	" 9/	" 9/	9/	9/	9/	9/	9/	9/		0		"
	t_{WHAX}		58	" 11/	" 11/	" 11/	11/	11/	11/	11/	11/	11/		0		"
	t_{WHDX}		59	" 11/	" 11/	" 11/	11/	11/	11/	11/	11/	11/		0		"
	t_{WHQX}		60	" 10/	" 10/	" 10/	10/	10/	10/	10/	10/	10/		0		"
	t_{AVQV}		61	" 11/	" 11/	" 11/	11/	11/	11/	11/	11/	11/			12/	"
	t_{ELQV}		62	" 11/	" 11/	" 11/	11/	11/	11/	11/	11/	11/			12/	"
	t_{EHQZ}		63	" 8/	" 8/	" 8/	8/	8/	8/	8/	8/	8/		0	12/	"
	t_{ELWH}		64	" 11/	" 11/	" 11/	11/	11/	11/	11/	11/	11/		12/		"
	t_{AVWH}		65	" →	" →	" →	→	→	→	→	→	→		12/		"
	t_{WLWH}		66	" "	" "	" "	"	"	"	"	"	"		12/		"
	t_{DVWH}		67	" "	" "	" "	"	"	"	"	"	"		12/		"
	t_{WLQZ}		68	" 10/	" 10/	" 10/	10/	10/	10/	10/	10/	10/		0	12/	"
10																
11																

8/ For these tests in subgroups 9, 10, and 11, the Z.PAT pattern (see table VI) shall be used. Input and output conditions appear in table IIIA. Timing diagrams appear in the write and read cycles of figure 5. Subgroups 9, 10, and 11 shall be performed with V_{CC} = 4.5 V and 5.5 V.

9/ For these tests in subgroups 9, 10, and 11, the MARCH pattern (see table VI) shall be used. Input and output conditions appear in table IIIA. Timing diagrams appear in the write and read cycles of figure 5. Subgroups 9, 10, and 11 shall be performed with V_{CC} = 4.5 V and 5.5 V.

10/ For these tests in subgroups 9, 10, and 11, the RMZPAT pattern (see table VI) shall be used. Input and output conditions appear in table IIIA. Timing diagrams appear in the write and read cycles of figure 5. Subgroups 9, 10, and 11 shall be performed with V_{CC} = 4.5 V and 5.5 V.

11/ For these tests in subgroups 9, 10, and 11, the LORDCB and ROMPAT patterns (see table VI) shall be used. Input and output conditions appear in table IIIA. Timing diagrams appear in the write and read cycles of figure 5. Subgroups 9, 10, and 11 shall be performed with V_{CC} = 4.5 V and 5.5 V.

12/

Parameter	Device type		
	01	02	03
t_{AVQV}	35	45	55
t_{ELQV}	35	45	55
t_{EHQZ}	15	20	25
t_{ELWH}	30	35	45
t_{AVWH}	30	35	45
t_{WLWH}	30	35	45
t_{DVWH}	15	20	25
t_{WLQZ}	15	15	20

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. When the alternate screening procedures for class B microcircuits of method 5004 of MIL-STD-883 is applied, the following additional items are applicable:

(1) Internal visual, method 2010 of MIL-STD-883, condition B. In addition to the changes indicated by the alternate screens of method 5004, the following shall apply:

Metallization inspection shall be applicable to the top layer metal conductor (i.e., Al) and need not include "underlying conductors" such as polysilicon.

b. Delete the sequence specified in 3.1.10 through 3.1.14 of method 5004 and substitute lines 1 through 7 of table II herein.

c. Burn-in test (method 1015 of MIL-STD-883).

(1) Dynamic test (test conditon D) using circuit shown on figure 4, or equivalent. Ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

d. Interim and final electrical test parameters shall be as specified in table II herein.

e. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S dynamic burn-in shall be 5 percent, based on the exact number of devices submitted to each separate burn-in.

b. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

c. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits which consist of tests specified in table IV herein or electrical parameter limits specified in table III, subgroup 2 are defective and shall be removed from the lot. The verified failure divided by the total number of devices in the lot initally submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510 and 4.3.1 herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.5).

4.3.1 Microcircuit qualification extension. When authorized by the qualifying activity, for qualification inspection, if a manufacturer qualifies to a faster access time device type which is manufactured identically to a slower access time device type on this specification, then the slower access time device type may be part I qualified without further qualification testing. At the manufacturer's request, the slower device types will be added to the QPL.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I, C_O measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Sample size is fifteen devices with no failures, and all input and output terminals tested. Capacitance shall be measured between the designated terminal and V_{CC} at a frequency of 1 MHz and a signal amplitude of 50 mV rms.

d. Dynamic switching tests shall be performed using the circuit configuration shown on figure 5.

e. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

f. Test vectors for static, dynamic, and functional tests are contained in table V which forms a part of the manufacturing test tape.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady-state life test circuits shall be conducted using test condition D and shall be submitted to the qualifying activity for approval.

b. Class S only, end-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 2 of group C inspection and shall consist of the tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 herein.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.2). When group E testing is performed, it shall be in accordance with table V of method 5005 of MIL-STD-883.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminals. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within +10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

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TABLE IIIA. Input and output conditions.
For table III subgroups 9, 10, and 11

Symbol	Termination(s)	A	B	Unit
V_{CC}	V_{CC}	4.5	5.5	V
V_{IH} 1/	Logic inputs	3.0	3.0	V
V_{IL} 1/	Logic inputs	0.0	0.0	V
V_{OH} 2/	Logic outputs	1.5	1.5	V
V_{OL} 2/	Logic outputs	1.5	1.5	V

1/ The values for input logic levels include the effects of added ground noise induced by the tester test head and load board.
2/ Output compare levels for subgroups 9, 10, and 11.
a. For V_{OH}: H > 1.5 V.
b. For V_{OL}: L < 1.5 V.

TABLE IV. Delta limits at +125°C.

Parameter 1/ 2/	Limits for all device types
I_{CCSB}	±300 µA

1/ Conditions and mechanizations of measurements shall be as specified in table III, subgroup 2.
2/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

TABLE V. Test vectors.

TABLE V, not printed herein, is a computer listing for use with table III herein. Each vector identifies activity states of the device. Table V, test vectors, shall be subject to approval by the qualifying activity.

NOTE: When the vectors are performed as specified in table III herein for subgroups 1, 2, 3, 7, 8, 9, 10, and 11, they verify the static, dynamic and functional, threshold and switching parameters at +25°C, -55°C and +125°C.

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TABLE VI. Functional algorithms.

Functional algorithms are test patterns which define the exact sequence of test used to verify proper operation of a random access memory (RAM). Each algorithm serves a specific purpose for the testing of the device. Each algorithm shall be performed in the exact sequence as described herein.

All patterns described are written to and read from the array in a topologically true manner.

For the functional tests of table III subgroups 7 and 8, seven test patterns are applied to the DUT at V_{CC} = 4.5 V, then V_{CC} = 5.5 V.

For the ac timing parameter verification of table III subgroups 9, 10, and 11, the patterns utilized are discussed in detail. These patterns are exercised first at V_{CC} = 4.5 V, then at V_{CC} = 5.5 V.

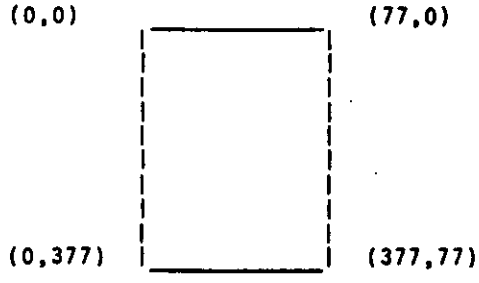
The seven patterns exercised during the functional testing shall be as follows:

LDRD0S
LDRD1S
LDRDCB
COLPAT
ROWPAT
MACPAT
SRNDST

Procedures for functional algorithms.

Steps that reference writing to or reading from "the entire array" begin address sequencing at the first address (0,0) and end at the last address (377,77). This last address in octal is equivalent to 256 rows by 64 columns in decimal (16,384 locations). That is, the addressing starts at the upper-lfet corner of the die and continues to the lower-right corner.

All patterns read data from the top cell to the bottom cell starting at column 0 and finishing at column 64 (0 to 77 octal).



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TABLE VI. Functional algorithms - Continued.

LDRD0S

Step 1 - Write zeros to the entire array.
Step 2 - Read zeros from the entire array.
Step 3 - Write ones to the entire array.
Step 4 - Read ones from the entire array.

LDRD1S

Step 1 - Write ones to the entire array.
Step 2 - Read ones from the entire array.
Step 3 - Write zeros to the entire array.
Step 4 - Read zeros from the entire array.

LDRDCB (Topologically true checkerboard)

Step 1 - Write checkerboard data to the entire array.
Step 2 - Read checkerboard data from the entire array.
Step 3 - Write checkerboard bar data to the entire array.
Step 4 - Read checkerboard bar data from the entire array.

Example:

Checkerboard	Checkerboard bar
0101	1010
1010	0101
0101	1010
1010	0101

COLPAT

Step 1 - Write column data to the entire array.
Step 2 - Read column data from the entire array.
Step 3 - Write column bar data to the entire array.
Step 4 - Read column bar data from the entire array.

Example:

Column	Column bar
0101	1010
0101	1010
0101	1010
0101	1010

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TABLE VI. Functional algorithms - Continued.

ROWPAT

Step 1 - Write row data to the entire array.
Step 2 - Read row data from the entire array.
Step 3 - Write row bar data to the entire array.
Step 4 - Read row bar data from the entire array.

Example:

Row	Row bar
0000	1111
1111	0000
0000	1111
1111	0000

MACPAT

Step 1 - Write zeros to the entire array.
Step 2 - Read first cell (test cell) 3 times.
Step 3 - Read the complement of the test cell 3 times; increment address.
Step 4 - Repeat steps 2 and 3 until the entire array has been checked.
Step 5 - Write ones to the entire array.
Step 6 - Read the test cell 3 times.
Step 7 - Read the complement of the test cell 3 times; increment address.
Step 8 - Repeat steps 6 and 7 until the entire array has been checked.

SRNDST (Please refer to the array below)

Step 1 - Write zeros to the entire array.
Step 2 - Write a one to cell 1 (test cell).
Step 3 - Read a one from cell 1.
Step 4 - Read a zero from cell 2.
Step 5 - Read a zero from cell 3.
Step 6 - Read a zero from cell 4.
Step 7 - Read a zero from cell 5.
Step 8 - Read a zero from cell 6.
Step 9 - Read a zero from cell 7.
Step 10 - Read a zero from cell 8.
Step 11 - Read a zero from cell 9.
Step 12 - Read a one from cell 1.
Step 13 - Write a zero to cell 1.
Step 14 - Read a zero from cell 1; increment address.
Step 15 - Repeat steps 2 through 14 until the entire array has been checked.

Example:

7	8	9
6	1	2
5	4	3

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TABLE VI. Functional algorithms - Continued.

NOTE: LDRDCB and ROWPAT patterns are discussed in the "functional test patterns" section.

Z. PAT

Step 1 - Write zeros to the entire array.
Step 2 - Read ones from the entire array.
Step 3 - Write ones to the entire array.
Step 4 - Read zeros from the entire array.

NOTE: Comparators levels are flipped in this test so that they actually check for opposite data.

RMZPAT

Step 1 - Write zeros to the entire array.
Step 2 - Read ones from the entire array.
Step 3 - Write ones to the entire array.
Step 4 - Read zeros from the entire array.

NOTE: Comparators levels are flipped in this test so that they actually check for opposite data. A read occurs after WE pulses low (i.e., within the write window.)

LDRCOL

Step 1 - Write column data to the entire array (XFAST).
Step 2 - Read column data from the entire array (XFAST).
Step 3 - Write column bar data to the entire array (XFAST).
Step 4 - Read column bar data from the entire array (XFAST).

NOTE: XFAST denotes the direction that data is written and read. This type of addressing scheme starts at cell 0,0 and continues across each row from row 0 to row 256.

MARCH

Step 1 - Write zeros to the entire array.
Step 2 - Read a zero from cell 1 (test cell).
Step 3 - Write test cell to one.
Step 4 - Read test cell; increment address.
Step 5 - Repeat steps 3 and 4 until the entire array has been checked.
Step 6 - Read a one from cell 1.
Step 7 - Write test cell to zero.
Step 8 - Read test cell; increment address.
Step 9 - Repeat steps 7 and 8 until the entire array has been checked.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of MIL-STD-883, method 5003), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by, or direct shipment to the Government.

h. Requirements for JAN marking.

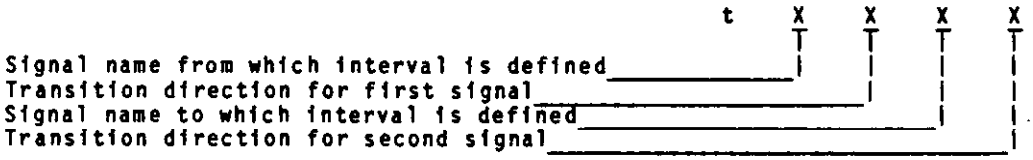
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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{SS}	Common or reference voltage mode.
V_{CC}	Supply voltage.
A0-A13	Address inputs, used to address 1 of 16,384/4 bit locations in static storage array.
$\overline{CS}$	Chip enable, used along with the write-enable ($\overline{WE}$) signal to control the state of the data-in and data-out bus (I/O1-I/O4).
I/O1-I/O4	Data input-output, 4 bit wide data bus through which data is stored or accessed.
$\overline{WE}$	Write-enable, used to select the Read or Write mode.
I_{DD}	Supply current.
I_{IH}, I_{IL}	Input leakage current.
I_{OZH}, I_{OZL}	High impedance output leakage current.
V_{IL}	Logical low input voltage.
V_{IH}	Logical high input voltage.
V_{OL}	Logical low output voltage.
V_{OH}	Logical high output voltage.
C_I	Input capacitance.
C_O	Output capacitance.
t_{ELQV}	Chip select access time.
t_{AVQV}	Address access time.
t_{EHQZ}	Chip select to output disable.
t_{AVQX}	Output hold after address change.
t_{ELQX}	Chip select to output active.
t_{WLWH}	Write pulse width.
t_{ELWH}	Chip select to end of write.
t_{DVWH}	Data setup to end of write.
t_{WHDX}	Data hold after end of write.
t_{AVWL}	Address setup before write enable low.
t_{AVWH}	Address valid to end of write.
t_{WHAX}	Write recovery time.
t_{WLQZ}	Write enable low to output in High-Z.
t_{WHQX}	Write enable high to data out, active time.
t_{PD}	Chip disable to power down time.
t_{PU}	Chip select to power up time.

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6.3.1 Timing parameter abbreviations. All timing abbreviations use lower case characters with subscripts. The initial character is always t and is followed by four descriptors. These characters specify two signal points arranged in a "from-to" sequence that define a timing interval. The two descriptors for each signal point specify the signal name and the signal transition. Thus the format is:



a. Signal definitions:

A = Address
D = Data in
Q = Data out
W = Write enable
E = Chip enable
O = Output enable

b. Transition definitions:

H = Transition to high
L = Transition to low
V = Transition to valid
X = Transition to invalid or don't care
Z = Transition to off (high impedance)

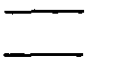
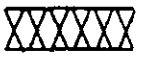
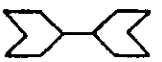
Example:

The example shows write pulse setup time defined as t_{WLEH} time from write enable low to chip enable high.

c. Timing limits: The table of timing values shows either a minimum or a maximum limit for each parameter. Input requirements are specified from the external system point of view. Thus, address set-up time is shown as a minimum since the system must supply at least that much time (even though most devices do not require it). On the other hand, responses from the memory are specified from the device point of view. Thus, the access time is shown as a maximum since the device always provides data prior to that time.

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d. Waveforms:

WAVEFORM SYMBOL	INPUT	OUTPUT
	MUST BE VALID	WILL BE STEADY
	CHANGE FROM H TO L	WILL CHANGE FROM H TO L
	CHANGE FROM L TO H	WILL CHANGE FROM L TO H
	DON'T CARE ANY CHANGE PERMITTED	CHANGING STATE UNKNOWN
	DOES NOT APPLY	CENTER LINES IS HIGH IMPEDANCE OFF STATE

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device B (see 1.2.2), and lead material and finish C (see 3.3). Longer length and lead forming shall not affect the part number.

6.5 Handling. MOS devices shall be handled with caution to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surfaces.

b. Ground test equipment, tools, and operator.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive foam or carriers.

e. The use of plastic, rubber, or silk in MOS areas should be avoided.

f. Relative humidity should be maintained above thirty percent.

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6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	51C98-35
02	51C98-45
03	51C98-55

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1050)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/613	MICROCIRCUITS, DIGITAL, VHSIC, CMOS, 65,536-BIT SELECTABLE MODE, STATIC RANDOM ACCESS MEMORY (SRAM) MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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